



Bangladesh University of Business and Technology

Course Title: Computer Networks Lab

Course Code: CSE 320

LAB REPORT

Experiment Name: RIP Routing Protocol

Experiment No.: 05

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Introduction

The Routing Information Protocol (**RIP**) is a widely used, simple distance-vector protocol designed for small networks, utilizing hop count as its metric with a **15-hop** limit. It uses the **Bellman-Ford algorithm**, exchanging full routing tables with neighbors every **30 seconds**.

In this experiment, we will implement RIPv1 (Classful) routing using a set of network devices (routers and switches) and client hardware.

Devices and Tools

- Cisco Packet Tracer v9.0.0
- 2960-24TT Switch × 6
- Router-PT × 3
- Client PC × 12

Client Addressing Table

Device	IP Address	Subnet Mask	Default Gateway
PC4	192.168.1.2	255.255.255.0	192.168.1.1
PC5	192.168.1.3	255.255.255.0	
PC6	192.168.2.2	255.255.255.0	192.168.2.1
PC7	192.168.2.3	255.255.255.0	
PC8	192.168.3.2	255.255.255.0	192.168.3.1
PC9	192.168.3.3	255.255.255.0	
PC10	192.168.4.2	255.255.255.0	192.168.4.1
PC11	192.168.4.3	255.255.255.0	
PC12	192.168.5.2	255.255.255.0	192.168.5.1
PC13	192.168.5.3	255.255.255.0	
PC14	192.168.6.2	255.255.255.0	192.168.6.1
PC15	192.168.6.3	255.255.255.0	

Router Interfacing Table

Device	FastEthernet0/0	FastEthernet1/0	Serial2/0	Serial3/0
Router2	192.168.1.1	192.168.2.1	192.168.7.1	N/A
Router3	192.168.3.1	192.168.4.1	192.168.7.2	192.168.8.1
Router4	192.168.5.1	192.168.6.1	192.168.8.2	N/A

Procedure

The following steps were meticulously followed to setup static routing among the network and client devices:

Preparing the Network:

- In the Cisco Packet Tracer application, we add three new **Routers (Router-PT)** and two child **Switches (2960-24TT)** for each of them. Then we add twelve Client PCs (PC4-PC15). We connect two clients with each of the switches. We connect the network devices using the **Auto Cabling** tool.
- We also interconnect **Router2, Router3, and Router4** using the **Auto Cabling** tool.

Configuring the client PCs:

- Single click the **PC4** and a new window appears.
- Under the **Desktop** tab, go to IP Configuration.
- Add the **IPv4 Address, Subnet Mask, and Default Gateway** according to the aforementioned **Client Addressing Table**. Repeat the steps for **PC5-PC11**.

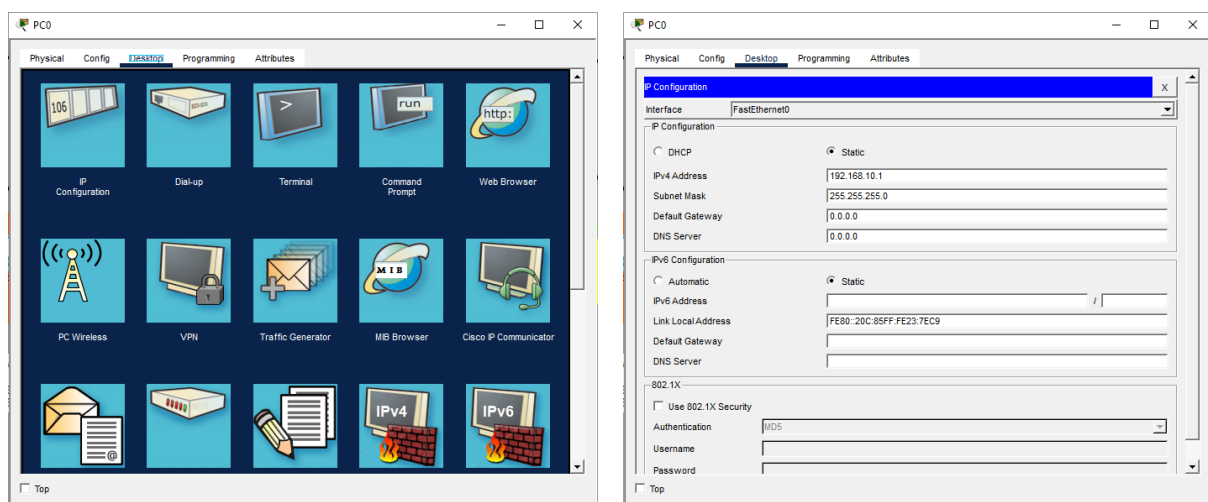


Fig 1.1: Configuring the client PCs.

Configuring the Routers:

- Click **Router2** and a new window will appear.
- Under the **Config** tab, we configure the IP addresses and Subnet masks of **FastEthernet0/0**, **FastEthernet1/0**, **Serial2/0**, and **Serial3/1** according to the aforementioned **Interfacing Table**. We make sure to turn on the respective ports.

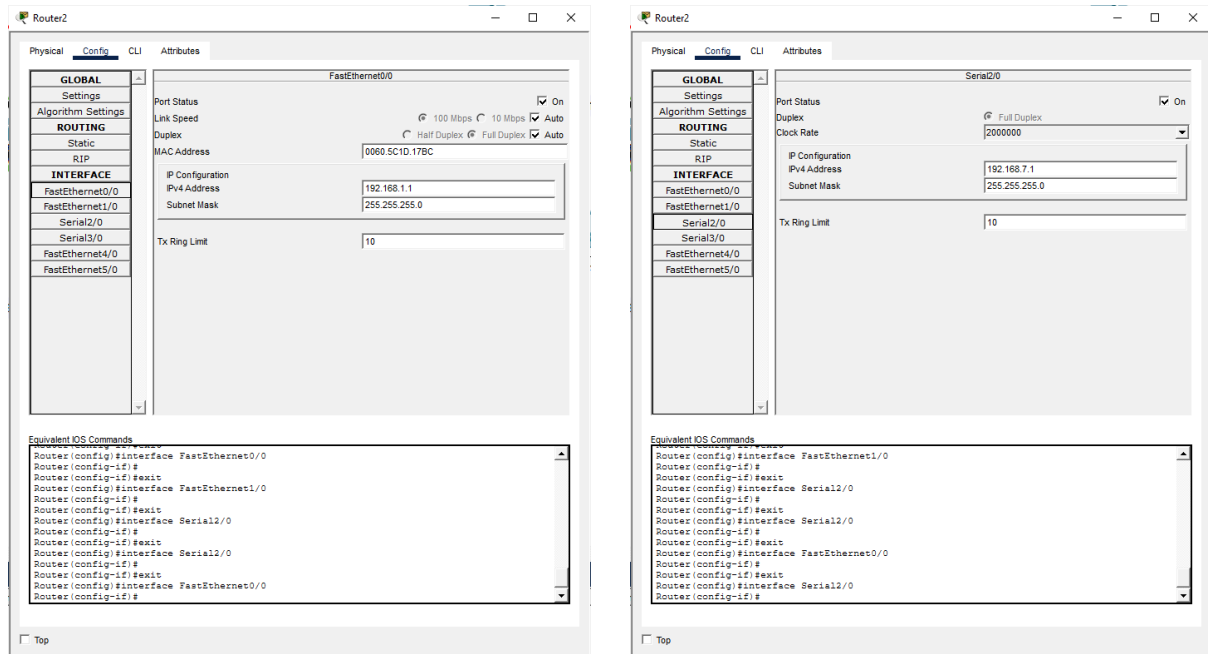


Fig 1.2: Configuring the Interfaces.

- We navigate to the **RIP** section and configure the routes. RIP routing is defined using network addresses.

For example: the Router2 has two subnets: 192.168.1.1-2 and its own IP address is 192.168.7.1.

Therefore we add –

- 192.168.1.0
- 192.168.2.0
- 192.168.7.0

In the RIP routing for **Router2**.

Router3 is connected (as a bridge) with **Router2** and **Router4**. It has two subnets: 192.168.1.3-4 and its network addresses are 192.168.7.2 and 192.168.8.1.

Therefore we add -

- 192.168.3.0
- 192.168.4.0
- 192.168.7.0
- 192.168.8.0

In the RIP routing for **Router3**.

Router4 is identical to **Router2** with hop addresses -

- 192.168.5.0
- 192.168.6.0
- 192.168.8.0

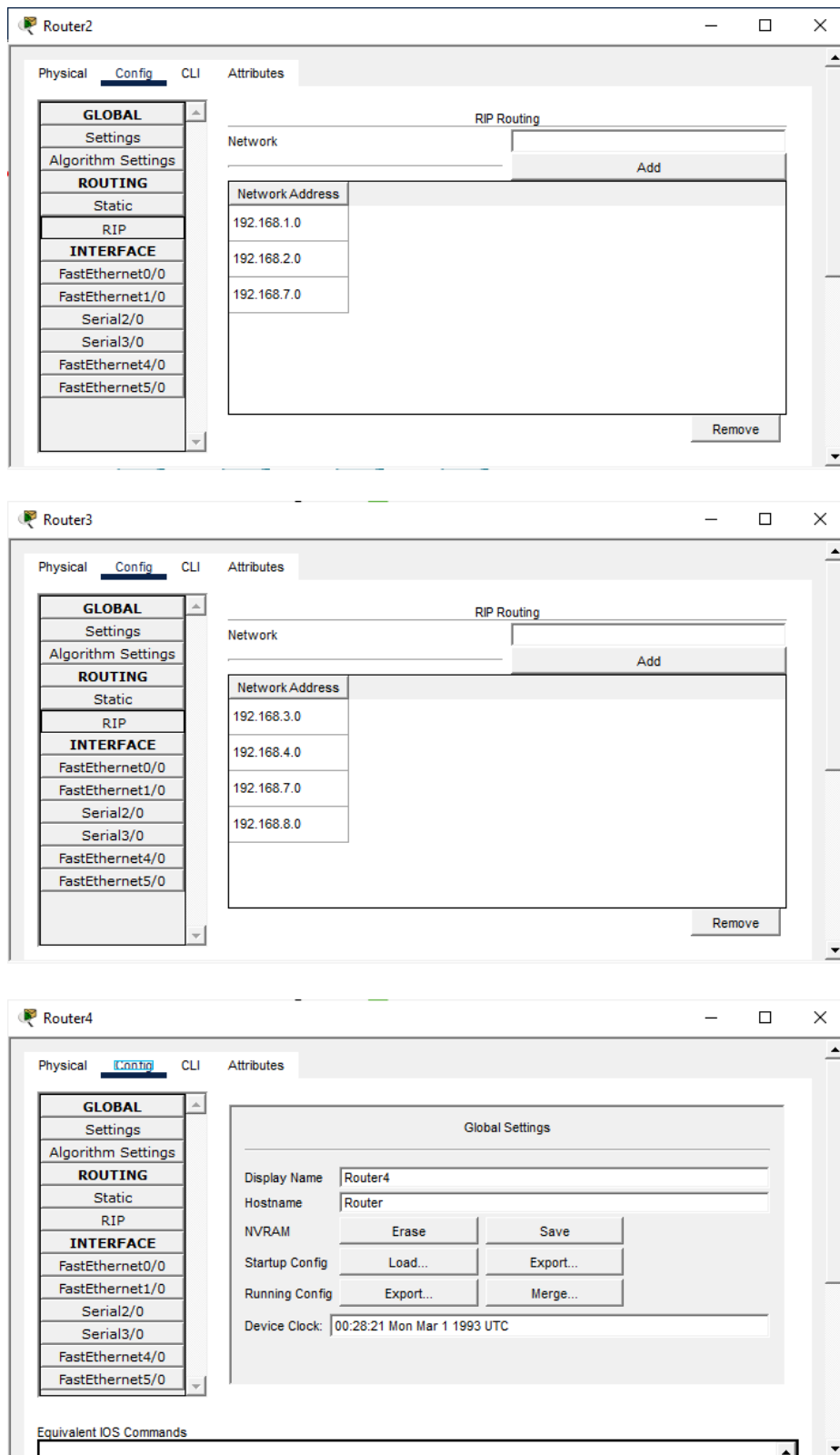


Fig 1.3: Configuring the RIP Routes.

- We repeat all the steps for Router3 and Router4.

Result

Once we have followed all the steps mentioned in the previous sections, we should have a working RIP routing configuration of three Routers, six Switches and twelve client PCs:

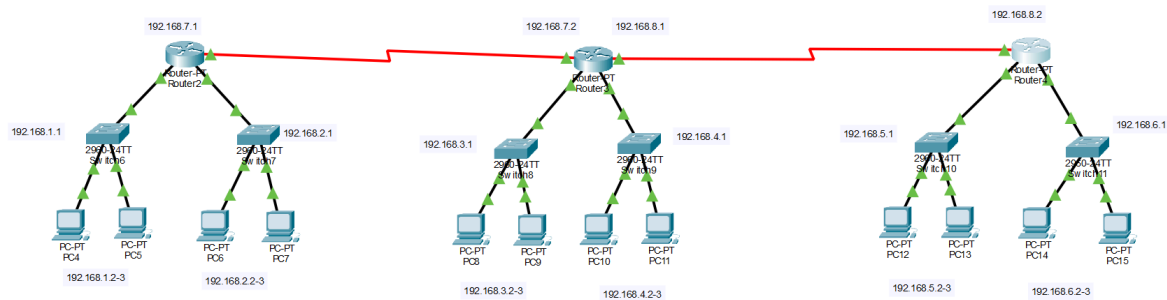


Fig 2.1: RIP Routing.

Testing and Verification

We send messages from client to client to check whether the data is being transmitted properly, ensuring that **RIP Routing** is properly implemented.

- From the toolbar, click “Add Simple PDU”. The cursor would turn into a message icon.
- Click on **PC4** and then under the same router, click **PC8**.
- If everything goes as expected, we should see a successful communication.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
...	Successful	PC4	PC8	ICMP	Green	0.000	N	0	(e...	
...	Successful	PC5	PC9	ICMP	Purple	0.000	N	1	(e...	
...	Successful	PC7	PC10	ICMP	Orange	0.000	N	2	(e...	
...	Successful	PC7	PC12	ICMP	Cyan	0.000	N	3	(e...	

Fig 3.1: Testing communication via RIP routing.

Conclusion

The objective of the fifth experiment was successfully achieved by configuring three different Routers, six Switches, and twelve client Devices to implement a network with RIP routing. It provided core skills for implementing RIP routing for non-scalable, easy to configure, small networks.
